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Metal-borohydrides as electrolytes for solid-state Li, Na, Mg and Ca batteries: a first-principles study

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ABSTRACT: Metal borohydrides are a family of materials that were recently discovered to have extraordinary ionic conductivities, making them promising candidates as electrolytes for solid state batteries (SSB). In fact, various groups have measured the ionic conductivities and assembled batteries using specific borohydrides. However, there are no comprehensive studies assessing the thermodynamic properties or discussing the suitability of metal borohydrides as electrolytes in SSBs, especially for beyond-lithium applications. In this work, we investigate the electrochemical stability, interfacial characteristics, mechanical properties, and ionic conductivities of Li, Na, Ca and Mg borohydrides using first principles calculations. Our results suggest that Li and Na borohydrides are unstable at high voltages. However, the corresponding decomposition products, i.e., $B_{12}H_{12}^{2-}$-containing phases, have wide electrochemical windows which protect the electrolyte, leading to large electrochemical windows as wide as 5 V. In addition, our simulations indicate that metal borohydrides are ductile, suggesting facile processing. However, their low shear moduli may result in metal dendrite formation. For Ca and Mg borohydrides, while they possess reasonably good electrochemical stability, the low cationic diffusivity may impede their practical use. Finally, the anion rotation barrier was shown to correlate with the superionic phase transition temperature, suggesting that anion mixing may be a potential approach to achieve room-temperature superionic conductivity.

1. Introduction

Solid-state batteries (SSBs) are among the most promising energy storage devices that may replace lithium ion batteries (LIBs). [1-6] This is because SSBs have, in principle, a higher energy density compared to LIBs. [3, 7] Additionally, SSBs are intrinsically safer because the flammable and volatile organic electrolytes of LIBs are replaced with relatively inert solids. [1, 3, 5] Although the chemistries of solid state and organic electrolytes are quite different, the basic requirements are similar, electrolytes need 1) to be electron insulating; 2) to have ionic conductivities large enough to enable high power output; 3) to be chemically compatible with electrode materials; 4) to have an electrochemical window wide enough to cover the working potential of the battery. Many families of solid electrolytes (SEs) have been studied, including oxides, [8-11] sulfides, [12-15] hydrides, [16, 17] halides, [18-22] and phosphates,[23-25] where each such family of materials has strengths and weaknesses. For example, sulfide-type $Li_{10}GeP_2S_{12}$ (LGPS) has an ionic conductivity comparable or even superior to that of organic liquid electrolytes. However, LGPS tends to be reduced by Li metal. [26, 27] Oxide-type $Li_7La_3Zr_2O_{12}$(LLZO) based-materials are less conductive than LGPS but their stability against Li is superior than that of sulfides.[26] Hydride-type $LiNH_2$ is mechanically flexible, allowing facile processing, and is stable against Li anodes but it is poorly compatible with most cathodes. [16] Materials in the halide-type antiperovskite family, such as Li_3OCl, have good conductivity but are very sensitive to the synthesis conditions. [18, 19, 21, 22]Finally, phosphates and borates, such as $LiTi_2(PO_4)_3$ and Li_3BO_3, are usually characterized by a relatively low conductivity. [23-25]

Recently, metal borohydrides have drawn considerable research interest because of the high Na and Li conductivity. [28-30] This family of materials has the composition $M_xB_yH_z$, where M are either alkali (Li, Na, K and Rb) or alkaline-earth (Mg, Ba, Ca, Sr and Ba) metals and B_yH_z are borohydride anions. In a typical metal borohydride crystal, these anions form a scaffold to hold the cations. **Figure 1 (a).** shows several commonly found borohydride anions, including, BH_4^-, $B_6H_6^{2-}$, $B_{10}H_{10}^{2-}$, and $B_{12}H_{12}^{2-}$.[17] For illustration purposes, we show the primitive cell of crystalline $Li_2B_{12}H_{12}$ in **Figure 1 (b)**, where the $B_{12}H_{12}^{2-}$ anions constitute a face-centered-cubic (FCC) framework and the Li^+ cations are accommodated within the spaces between the anions. [32]

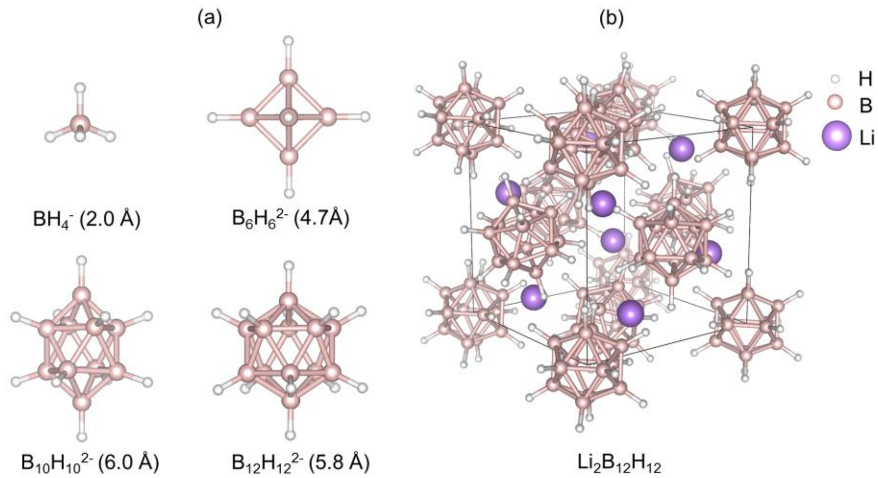

Figure 1. (a) Structure of typical borohydride anions. The sizes are given for reference and are defined as the greatest H-H distance within the anion cluster.[31] (b) The structure of the $Li_2B_{12}H_{12}$ crystal.

At low temperature, the anions are usually ordered. They vibrate around their nominal lattice site, leading to low cation diffusivity. [17] Instead, when the temperature increases, the anions rotate and their orientations are disordered, leading to both orientational anionic disorder and to superionic conduction.[17]

Metal borohydrides were recognized as hydrogen storage materials as early as 1953 and have been widely investigated since. [33] [17, 34, 35] In 2006, Nakamori *et al.* discovered that $LiBH_4$ has a Li conductivity as high as 10^{-3} S cm^{-1} at 390 K.[36, 37] Several groups have since tested $LiBH_4$ in full batteries. [38-40]Following these pioneering works, other metal borohydrides were investigated as SEs. In 2013, Buckley *et al*. and Teprovich *et al*. studied the phase transition and ion conduction in $Li_2B_{12}H_{12}$. [41, 42] T. J. Udovic and co-workers substituted $B_{12}H_{12}^{2-}$ with other anions and obtained $Li_2B_{10}H_{10}$, $LiCB_{11}H_{12}$, and $LiCB_9H_{10}$.[30, 43,29] These materials have an ionic conductivity as high as 0.03 S cm^{-1} at 354 K. [29] Interestingly, Na-conducting borohydrides ($Na_2B_{12}H_{12}$, $Na_2B_{10}H_{10}$, $NaCB_{11}H_{12}$ and $NaCB_9H_{10}$) exhibit even higher conductivities and, to the best of our knowledge, $NaCB_9H_{10}$ currently holds the record for the highest Na conductivity with 0.03 S cm^{-1} at 297 K. [28-30] Despite the aforementioned achievements, no existing borohydride has a conductivity greater than 0.01 S cm^{-1} at RT. To achieve the desired RT conductivity, two approaches could be taken. One is to lower the phase transition temperature because it is known that borohydrides have high ionic conductivities in their disordered phase. Possible methods could be substituting the anionic groups with halides or by high intensity mechanical ball milling.[44] [45] [46] Another way of tackling the low RT conductivity problem is to enhance the intrinsic cation diffusivity, which is the product of two factors: 1) the concentration of charge carriers, which is related to the cation vacancy formation energy; and 2) the intrinsic jump rates of charge carriers, which is the cationic diffusion barrier.[47] [48] Both have been used to improve the conductivity in borohydrides. For example, Takamura and co-workers increased the conductivity of $LiBH_4$ by doping Ca on the Li site, which promotes the Li vacancy concentration. [49] Conversely, Remhof *et al.* reduced the Li migration barrier and enhanced the Li diffusion in $LiBH_4$ by substituting BH_4^- anions with NH_2^-. [50]

Due to the fast ion transport in metal borohydrides, a number of groups have studied the structural properties and the mechanism of diffusion. The disorder phase transition was first correlated to superionic behavior for $LiBH_4$ by Orimo *et al*. [36] [37] Later, Orimo *et al.*[51] carried out ab-initio molecular dynamics simulations and explained the origin of fast Li conduction. They found that the fast Li diffusion is related to the atom occupation splitting, where the fully occupied Li sites split into two partially occupied sites above the phase transition temperature. In 2014, Buckley *et al.* found that the $Li_2B_{12}H_{12}$, a type of Li borohydride containing larger anions, also undergoes a similar phase transition.[41] Udovic and co-workers investigated the correlation between cation transport and the anion disorder using $Na_2B_{12}H_{12}$ as a model material. [52] The same group carried out neutron powder diffraction and nuclear magnetic resonance on several Na-Li based borohydrides and derived the structural and dynamical properties. [43, 53, 54] Despite these pioneering studies on the correlation between cationic diffusion and anionic rotations, many other properties need to be known. In particular, it is critical to assess the electrochemical stability and mechanical properties of these materials (i.e. bulk and shear moduli).

In this article, we present the chemical and electrochemical stability, the interfacial reactivity, and the mechanical strength of several Li, Na, Ca, and Mg conducting metal borohydrides. After introducing the computational methods and the materials systems, we discuss the computed compositional phase diagrams to assess the intrinsic thermodynamic and chemical stability (against metal anodes) of the materials studied. We also evaluated the electrochemical stability of metal borohydrides under applied potential based on the band structure and grand-potential phase diagrams. Furthermore, we investigated the mechanical characteristics of the metal borohydrides by calculating the elastic constants. The computed Pugh ratio, which is a descriptor for the ductility of a material, is used to infer its mechanical processability. [55] [56] On the other hand, the shear modulus can be linked to the capability of the material to prevent the formation of dendrites. [57] Additionally, we attempted to assess the charge carrier concentration, a quantity directly relevant to the cationic conductivity, by calculating the cationic vacancy formation energies. Finally, since a low superionic phase transition temperature is critical for the practical use of these material at room temperature, we correlated the DFT-calculated anion rotation barriers to experimental phase transition temperatures and provide strategies to lower the phase transition temperature, i.e., by making solid solutions using phases with different anion sizes.

2. Computational method

2.1 Ground state energies and electronic structures

The ground state energies of some phases were already computed as part of the Materials Project (MP), [58] for this reason we directly retrieved these values from the MP database. As it is described in the materials systems and crystal structure section 2.2, we also considered other potential phases, for which we calculated the ground state energies using the same computational settings of the MP. All DFT calculations were done using VASP with the GGA-PBE functional. [59, 60] All calculations were spin-polarized except for molecular dynamics simulations.[59, 60] For the electron part, the energy cutoff for plane-waves was set at 520 eV. To model the core electrons, we used the Projector Augmented Wave (PAW) method.[61] The Brillouin zone was sampled on a Monkhorst-Pack k-point mesh with density of 1000/(number of atoms in the cell).[62] For the ionic relaxation, all degrees of freedom including the lattice parameters were allowed to change until the energies converged to 1 meV/atom. The computational scheme was further evaluated by relaxing the structure of $LiBH_4$ with several different exchange correlation functionals, i.e., LDA, GGA-PBE, GGA-PBEsol and GGA-PBE+D3. [63] [64] The results are shown in **Table. S1**. All calculations yielded lattice constants in reasonable agreement with the experimental results. We chose to use GGA-PBE functional for consistency with the MP database.

The calculation of the density of states of the bulk systems was carried out using the MBJ meta-GGA functional (modified Becke-Johnson exchange potential in combination with LDA-correlation functional).[65, 66] This method was chosen because it yields band gaps with an accuracy similar to those obtained using hybrid functional or GW methods but with a computational cost comparable to semi-local DFT calculations. [66, 67]

2.2 Material systems and crystal structures

We directly retrieved the relaxed structures and the corresponding energies of the phases that are already included in the MP database. We also considered other potential phases within the M-B-H (M=Li, Na, Ca, Mg) ternary composition range. In particular, we included all the phases with composition $A_xB_nH_n$ (A=Li, Na; x=1, 2; n=11, 12) and MB_nH_n (M=Ca, Mg; n=11, 12) that were previously reported in literature. [17] In order to search for low energy structures, we also considered cation-exchanged and anion-rotated polymorphs. For example, we substituted all of the Li atoms in the known Li_3BH_6 crystal to Na and included this entry when building the Na-B-H phase diagrams. Phases we added to the MP database in this calculation and their references are listed in part 1 section 1 of the supplementary information (SI).

2.3 Phase diagrams, electrochemical windows and vacancy formation energies

The compositional phase diagrams were constructed using the pymatgen code based on DFT ground state energies (0 K and 0 atm). [68, 69] The detailed description of the algorithms used to construct the phase diagrams are described elsewhere. [70, 71] The formation energies of all phases of interest were calculated based on DFT and normalized by the number of atoms in the formula. Using the normalized formation energies, we searched for the convex hulls in the compositional space. To read the phase diagram, one needs to specify the composition of interest and calculate its coordinate in the compositional space using lever rule. Once the coordinate is located, one can determine whether it is a stable phase or a mixed phase. The stable phases are shown using the black dots while the components of the mixed phases are given by the vertices of the triangle bounding that composition. For example, if one is interested in determining the phase of $LiB_{10}H_9$ (indicated as α in **Figure 2 (a)**), one sees from the diagram that it is be a mixture of B, B_9H_{11} and $Li_2B_{12}H_{12}$.

In order to obtain the thermodynamics of the metal borohydrides open to a reservoir of the corresponding metal, we also constructed the grand potential phase diagrams using the methodology of Ceder *et al.*[70] In these phase diagrams, the chemical potential μ_M of M (M = Li, Na, Ca and Mg) is given as a function of the electrode potential ϕ by

$$\mu_M(\phi) = \mu_M^0 - e\phi \qquad (1)$$

where μ_M^0 is the electrochemical potential of M and ϕ is the applied potential referenced to M.

The electrochemical window of a specific phase was calculated by constructing the grand potential phase diagram versus ϕ. The range of potentials ϕ within which the specific phase is stable (the phase is a vertex of the convex hull), is the electrochemical window of the phase. We should point out that the current approach does not take into account the kinetics effect. [72]

The cation vacancy formation energies were estimated using two methods based on charge-neutral configurations. First, similarly to the work of Miara *et al.*, [73] we considered the phase equilibria using the calculated phase diagram.

For example, for $LiBH_4$, we created a vacancy containing structure with the composition $Li_{7/8}BH_4$ (or $Li_7B_8H_{32}$ where one vacant Li site is present in the structure). Then the coordination of this phase on the phase diagram was calculated using the lever rule and shown in **Figure 2 (a)** using a red triangle. Since this structure is not on the convex hull, it will decompose into corresponding stable phases, i.e., BH_3, $LiBH_4$, and H_2 (the three vertex of the triangle covering $Li_{7/8}BH_4$). Therefore, the following reaction is obtained:

$$56\ LiBH_4 + 8\ BH_3 + 4\ H_2\ (g) \rightarrow 8\ Li_7B_8H_{32}, \quad \Delta E_{equilibrium} = 21.43\ eV \quad (2)$$

where $\Delta E_{equilibrium}$ is the reaction energy calculated based on phase equilibrium. The final vacancy formation energy was normalized by the number of Li atoms in the non-defective supercells, leading to $E_{vacancy-equilibrium} = \Delta E_{equilibrium}/64 = 0.33$ eV. Using this method, the BH_4^- anion is predicted to decompose. This reaction may be kinetically slow at room temperature as explained in Electrochemical Stability section 3.2. Therefore, we also considered a second approach, where a Li cation is directly extracted from the materials (without the decomposition of anions) following the reaction:

$$8\ LiBH_4 \rightarrow Li + Li_7B_8H_{32}, \quad \Delta E_{extract} = 4.70\ eV \quad (3)$$

This leads to a higher vacancy formation energy of $E_{vacancy-extract} = \Delta E_{extract}/8 = 0.59$ eV, which is higher than the previous method. This is because the metal borohydrides are unstable with the reaction product, i.e., lithium metal based on later analysis.

2.4 Elastic constants

To obtain the mechanical properties, we calculated the 6x6 elastic tensors (C_{ij}) by enforcing six finite distortions (0.01 Å) on the relaxed lattice and derived the elastic constants from the linear strain-stress relationship using VASP.[74] To estimate the mechanical properties of polycrystalline materials, we used the Voigt (V) and Reuss (R) models to calculate the bulk (B) and shear (G) moduli. This procedure gave us upper and lower bounds.[75] Details are presented in the part 2 section 1 of SI. We also calculated the Voigt-Reuss-Hill (VRH) modulus by averaging the Voigt and Reuss models.[76] The Pugh ratio was determined by dividing the bulk (B) and shear (G) moduli under the Voigt-Reuss-Hill limit. In all elastic property calculations, we used higher energy cutoffs of 700 eV and doubled the k-points on each direction compared with the ground state energy calculations.

2.5 Molecular dynamics simulations

We studied $CaB_{12}H_{12}$ and $MgB_{12}H_{12}$ using Born-Oppenheimer molecular dynamics (MD) simulations at desired temperatures using a Nose-Hoover thermostat with a fixed simulation box size (NVT ensemble). [77] The supercells were constructed so that at least 8 cations are present in the simulation box. The time steps were chosen to be 1 fs. All electron self-consistent calculations were carried out using VASP with an energy cutoff of 350 eV and an energy convergence criterion of 10^{-4} eV. The plane waves were sampled at the Gamma point only.

3. Results and discussion

3.1 Phase stability

The phase stability of metal borohydrides was studied by constructing A-B-H (A= Li, Na, Ca and Mg) ternary phase diagrams. For lithium, we found most of the phases reported in previous experimental work to be either thermodynamically stable or thermodynamically meta-stable. As shown in **Figure 2 (a)**, $LiBH_4$ and $Li_2B_{12}H_{12}$ lie on the convex hull, meaning that they are intrinsically stable. This is in agreement with the reports on the stability of the latter phase, where $Li_2B_{12}H_{12}$ was confirmed as an intermediate species during the hydrogen desorption of $LiBH_4$ via in-situ thermogravimetry, X-ray diffraction, and DFT calculations. [78, 32, 79] Several structures based on anion substitutions to $Li_2B_{12}H_{12}$ have also been recently reported. For example, Udovic *et al*. synthesized $Li_2B_{10}H_{10}$ and analyzed its crystal structure using neutron powder diffraction. [43] Interestingly, from our calculations, $Li_2B_{10}H_{10}$ is thermodynamically unstable at 0 K. As shown in **Table S2**, its energy is 53 meV/atom above the hull. Since this value is relatively small, such phase could be in principle stabilized by entropy at RT. Our results also suggest that other anion modifications may lead to meta-stable phases, such as $LiB_{10}H_8$ and LiB_3H_8, which have not been found experimentally. Apart from the intrinsic stability, the phase diagram also gives information on the chemical stability, i.e., the stability of a material when it is in contact with the corresponding metal anode. Generally, if there is no additional panel in the convex hull between two phases, those can coexist. As shown in **Figure 2 (a)**, all of the calculated Li borohydrides are thermodynamically unstable with lithium metal. For example, $LiBH_4$ tends to decompose into $Li_2B_{12}H_{12}$, LiH, and LiB depending on the Li-$LiBH_4$ ratio. Such electrolyte-anode instability may be detrimental to a solid state battery. Additionally, this finding is in contrast with previous experimental results where Li-borohydrides were found to be electrochemically stable with Li metal. [17,39, 40, 42] These results suggest that either the reactions are kinetically slow or that the interface is stabilized during the battery operation. This point will be discussed in the next section.

The phase stability of Na borohydrides is similar to that of Li borohydrides in spite of Na's larger ionic radius, see **Figure 2 (b)**. In both cases (A= Li, Na), ABH_4 and $A_2B_{12}H_{12}$ are thermodynamically stable while the substitutions of anions to others, such as $B_{10}H_{10}^{2-}$, leads to meta-stable phases. However, we should note that although the phase diagrams share some resemblance, the crystal structures (cations/anions packing) are usually different. For example, in $Li_2B_{12}H_{12}$, the anions are cubic-close-packed (ccp)/faced centered cubic (fcc) while in the Na counterpart ($Na_2B_{12}H_{12}$) they are hexagonal-close-packed (hcp). Regarding the chemical stability, our calculations suggest that, in contrast with their Li counterparts, $NaBH_4$ and $Na_2B_{12}H_{12}$ are stable against Na metal. Such stability might indicate that interfacial issues arise when assembling Na SSBs.

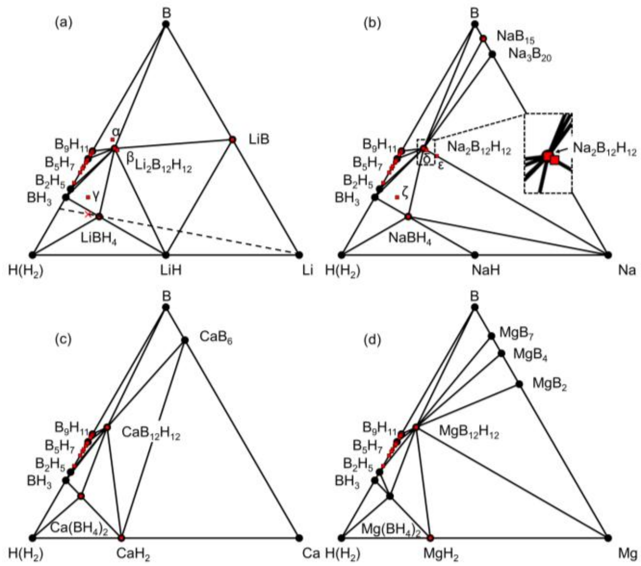

Figure 2. Compositional phase diagrams of (a) Li-B-H, (b) Na-B-H, (c) Ca-B-H and (d) Mg-B-H compositional phase diagrams. The calculated stable phases are indicated by the black dots with their compositions labeled with black fonts correspondingly. We also show possible metastable phases whose ground state energies are < 80 meV/atom above the hull. These phases are indicated by the red squares. Due to space limitation, their compositions are not shown in the Figure but captioned here: (α) $LiB_{10}H_9$, (β) $Li_2B_{10}H_{10}$, (γ) LiB_3H_8, (δ) $Na_2B_{10}H_{10}$, (ε) $Na_2B_6H_6$, and (ζ) NaB_3H_8. The dashed line in (a) represents the $LiBH_4$-Li compositional space. The red cross in (a) denotes the vacancy containing structure $Li_{7/8}BH_4$ (or $Li_7B_8H_{32}$).

Figure 2 (c) and **(d)** show the ternary Ca-B-H and Mg-B-H phase diagrams. For Ca and Mg, we found considerably fewer meta-stable phases. This might be due to the lack of input structures despite we collected, to the best of our knowledge, all known Ca and Mg borohydrides structures from literature and tried to derive several potential structures from the Li and Na counterparts as shown in **Figure S4** of the SI. For the considered cases, the BH_4^- and $B_{12}H_{12}^{2-}$ containing phases tend to be stable. Both $CaBH_4$ and $MgBH_4$ decompose into compounds containing $B_{12}H_{12}^{2-}$ when they are in contact with the metal anodes. Among all calculated Ca and Mg borohydride, $MgB_{12}H_{12}$ is the only thermodynamically stable phase against its metal anode.

Interestingly, the phase diagrams also explain the thermodynamic origin of the experimentally observed anion disorder phase transitions. For many of the compositions studied, various structures with identical phase diagram coordinates have very similar energies. An example is shown in the inset of **Figure 2 (b)**, where for the composition $Na_2B_{12}H_{12}$ (denoted with the arrow) a single stable phase is observed. Additionally, we also identified meta-stable phases with the same composition (denoted with the red square) which are < 80 meV/atom above the hull. In one of these structures, the ion arrangements are very close to the most stable ones but the $B_{12}H_{12}^{2-}$ anionic groups have slightly different orientations, see **Figure S4.** This finding supports the previous report on the anion disorder phase transitions in metal borohydride since such a small energy difference may imply that the meta-stable states are possible at finite temperatures. According to Udovic *et al.*, $Na_2B_{12}H_{12}$ goes through an anion disorder phase transition at elevated temperatures accompanied by a significant increase in Na conductivity. [52] Since the rotation of the $B_{12}H_{12}^{2-}$groups does not lead to significantly higher energies, such a process could be activated by thermal fluctuations. For $MgB_{12}H_{12}$ and $CaB_{12}H_{12}$, such phase transitions have not been reported experimentally. However, since we computed meta-stable phases for $MgB_{12}H_{12}$ and $CaB_{12}H_{12}$, we can speculate that, analogous with Li and Na borohydrides, a similar anion disordered phase transition may occur in $MgB_{12}H_{12}$ and $CaB_{12}H_{12}$. We examined such hypothesis, using *ab-initio* MD simulations, which showed that the $B_{12}H_{12}^{2-}$ anions in $CaB_{12}H_{12}$ rotate at 1500 K but not at 1000 K. The corresponding trajectories are shown in **Figure S3**.

3.2 Electrochemical stability

One critical requirement of an electrolyte is a wide electrochemical window, which allows the use of high-voltage cathodes. While a number of SEs, including LLZO and LGPS, have been reported to have wide electrochemical windows,[12, 80] recent studies combining cyclic voltammetry and DFT calculations show that some of these materials are intrinsically unstable as high-voltage cathodes. [26] For example, LLZO reacts with the high-voltage $LiMn_{1.5}Ni_{0.5}O_4$ and the battery stops after a short discharge. [26, 81, 82] This puts into question whether the novel metal borohydride-based SSB could be cycled at high-voltages. We computationally address this issue in two ways: 1) we calculated the band structures of the metal borohydrides; 2) we constructed the grand potential phase diagrams. With the former methods, we obtain an upper bound of the electrochemical stability window against corresponding inert electrodes, in contrast with the latter approach, for which a lower bound is obtained. [26, 83]

In the former method, the electrochemical window is interpreted as the voltage range within which the material is neither reduced nor oxidized. In principle, to oxidize a material, an electron needs to be removed from the highest occupied molecular orbital (HOMO) or the valence band maximum (VBM). To reduce a material, an electron needs to be added to the lowest unoccupied molecular orbital (LUMO) or the conduction band minimum (CBM).

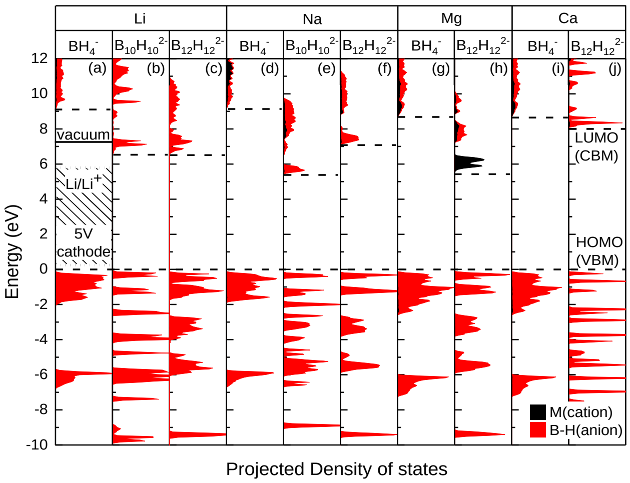

Figure 3. Partial density of states of M (M= Li, Na, Mg, and Ca) and B-H anion groups of related stable/meta-stable phases. All the energies are referenced to the corresponding Fermi level of the borohydrides. The VBMs and CBMs were denoted with dashed lines. For $LiBH_4$, the working potential range of the 5-V Li|$LiBH_4$|cathode battery is shown as the shaded area. The Li/Li$^+$ level is absolute Li electrode potential and the VBM of 5V-cathode (dotted line) is calculated by subtracting 5 eV from that. Detailed theory and computation method can be found in part 4 section 1 of SI and also previous work [84]

Thus, it is natural to correlate the electrochemical window of a material to its band gap, i.e., the gap between LUMO and HOMO. However, since it is non-trivial to reference the energy levels of different materials based on periodic DFT calculations and no chemical contact with the electrode material is considered, the results of this method, i.e., the band gap, should be regarded as an upper bound of the intrinsic electrochemical window against inert electrodes.[83] The partial density of states of the metal borohydrides are shown in **Figure 3**. The band gaps of all materials exceed 5 eV based on meta-GGA level calculations. This indicates a large electrochemical window against inert electrodes. In all the calculated entries, the lowest occupied states are always dominated by the anion species. It means that the anionic groups are likely the first part of the electrolyte to be oxidized under high voltages. This is in agreement with the predicted reactions from the grand-potential phase diagram, **Figure S6**, where the BH_4^- and $B_{12}H_{12}^{2-}$ anions are oxidized to B_xH_y species, e.g. B_2H_6 and B_2H_5, at high voltages. As mentioned above, the problem of this method is the alignment of band edges (determination of absolute energy of HOMO and LUMO). This could in general be solved by building a slab model and reference all energies to vacuum. [84-86] However, due to the high computational cost of surface-type models, we only reformed this adjustments for $LiBH_4$ (the implementative details are provided in part 4 section 1 of SI). As shown in **Figure 3**, the Li/Li$^+$ level (Li Fermi level with respect to vacuum) and the 5V-cathode level (Li Fermi level – 5 eV) are between HOMO (VBM) and LUMO (CBM) of $LiBH_4$. This indicates that $LiBH_4$ should be stable upon reduction of Li and should be compatible with high oxidative potentials. However, such result only provides a stability upper bound. In fact, the VBM of a high voltage 5V-cathode is very close to the VBM of $LiBH_4$ meaning the $LiBH_4$ might still be oxidized by the cathode and further analysis is required.

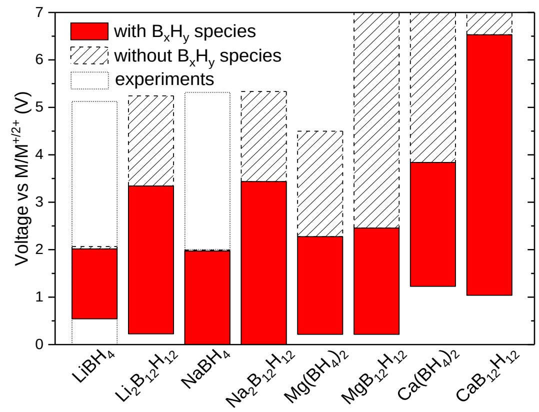

Figure 4. Thermodynamic electrochemical windows calculated from the grand-potential phase diagrams. The red area indicates the electrochemical window calculated with the B_xH_y species included and the shaded area indicates the ones calculated without the B_xH_y species included. The experimental values were obtained from the work of Unemoto *et al.* [39] and the review by Mohtadi *et al.*[17]

We then evaluated the electrochemical stability of the materials by constructing grand-potential phase diagrams. This method was previously applied to several oxide and sulfides and some of the predictions were later verified experimentally.[26, 87-89] **Figure 4** summarizes the electrochemical windows computed using this method. For Li and Na, we predicted that the BH_4^--containing phases are oxidized at ~2V vs Li/Li$^+$ (Na/Na$^+$). This is in contrast with the experimental cyclic voltammetry results suggesting their stability up to 5V. [39] Since many works reported that the decomposition of $LiBH_4$ is kinetically slow and the reaction only happens at elevated temperatures, [17] it is possible that our underestimated electrochemical window is due to the sluggish anion decomposition. [72, 90, 91] As mentioned previously, the electrochemical windows were estimated based solely on thermodynamics while such reactions might be so slow that the current could be hardly measured using CV at RT. [26, 72, 92] Following this, we analyzed the effect of discarding all B_xH_y phases, the products of the potential slow anion decomposition reactions, from the grand potential phase diagrams (**Figure 5 (a)** and **(b)**). [72, 93] However, the electrochemical windows of $LiBH_4$ and $NaBH_4$ are still much narrower than those reported experimentally (see in **Figure 4**). This indicates

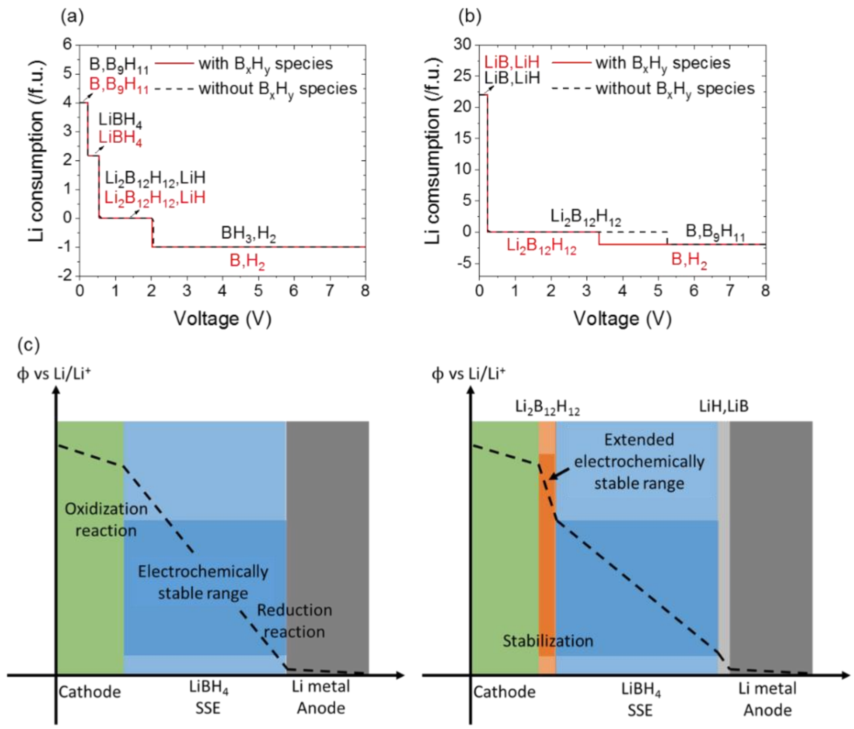

Figure 5. Decomposition products of (a) $LiBH_4$ and (b) $Li_2B_{12}H_{12}$ at different voltages vs Li^+/Li. For Na, the decomposition products are shown in **Figure S5**. (c) Illustration of the electrochemical stabilization of the interface.

the possibility of an alternative stabilization mechanism. Interestingly, we found that the decomposition products at high voltages are $Li_2B_{12}H_{12}$ and $Na_2B_{12}H_{12}$ (shown in **Figure 5 (a)-(b)** and **Figure S5**), two materials that were recently found to be ionically conductive. [42, 94] From **Figure 4**, the electrochemical windows of these two phases are much wider than their BH_4^- containing counterparts. Based on this, we propose the stabilization mechanism shown in **Figure 5 (c)**. The electrolyte, e.g., $LiBH_4$ and $Li_2B_{10}H_{10}$ initially reacts with the cathode material to form ionically-conductive $B_{12}H_{12}^{2-}$ containing interphases. Due to the electrochemical stability of the interphases, this thin layer protects the electrolyte and prevents the cathode materials to be further consumed by the reaction. This is in agreement with recent experiments by Orimo *et al.* who suggested that $Li_2B_{12}H_{12}$ forms between the cathode and the $LiBH_4$ electrolyte. [40] On the anode side, we predicted that both Li and Na borohydrides are likely to decompose to borides and hydrides at reductive potential (low voltages). A similar stabilization mechanism for the cathode | electrolyte interface may also apply to the anode side as shown in **Figure 5 (c)** where the electrolyte is protected by the binary compounds, such as LiB and LiH.

We want to note, however, that due to the low reduction potential (<0.25 V), the driving forces might be relatively small so such reaction might be slow or even hindered. We can draw some similarity with the LLZO|Li interface. Based on DFT calculations, Mo *et al.* reported that LLZO is also thermodynamically unstable against a Li anode. However, the reduction potential of LLZO (0.05 V) is close to that of Li deposition (0 V) with a correspondingly small thermodynamic force driving the reduction.[26] Therefore, it is not surprising that the LLZO|Li interface is found stable experimentally.

Recently, Ca and Mg batteries have received increased attention as alternatives to the lithium-based battery because of their high volumetric capacity and improved safety where no dendrites form. [95, 96] Thus, the electrochemical windows of Mg and Ca borohydrides were also evaluated. As shown in **Figure 4**, these phases exhibit similar stability compared to their Li/Na counterparts. At low potentials, they tend to be reduced by metallic anodes to form borides and hydrides. For Mg, the reduction potential is relatively low (<0.5 eV) indicating a relatively small driving force for the reaction. This might be the reason for the recently identified compatibility between the Mg anode and $Mg(BH_4)_2$/TMF electrolytes. [97] While we are not aware of any electrochemical experiments of Ca borohydrides, our calculations predict that the Ca borohydrides are severely reduced at the Ca anode with reaction energies of (~3.868 eV/f.u. for $Ca(BH_4)_2$). Experiments are however needed to assess the Ca|Ca borohydride interfacial problems. At high potentials, both Ca and Mg borohydrides possess relatively good stabilities against electrochemical oxidation. Considering the reactivity of conventional liquid Mg and Ca electrolytes with cathodes, a very thin layer of corresponding borohydrides may work as a protective coating layer for high-voltage (> 5 V) cathodes due to their electrochemical stability. [96]

3.3 Mechanical properties

One major advantage of SEs over their conventional organic counterparts is the ability to suppress metal dendrite growth thanks to the superior mechanical properties. While this statement holds true for most ceramic-type oxides, e.g., LLZO, [56] whether metal borohydrides are mechanically compatible with metal anodes is still an open question. Another critical property for solid-state-electrolytes is their ductility. For example, oxide SSEs are known to be brittle and likely break during mechanical loading. Specialized techniques need to be developed to process such thin layers.[5] We addressed these issues using DFT calculations.

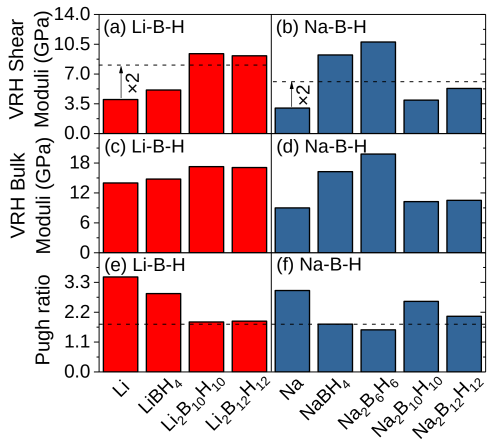

Figure 6. Panels (a) and (b) shear moduli, panels (c) and (d) bulk moduli, and panels (e) and (f) Pugh ratios of Li/Na metal and metal borohydrides, respectively. The shaded area in (a) and (b) indicates a doubled shear moduli of calculated Li and Na metal which indicates the lower limit which the electrolyte could resist dendrite formation.

We calculated the shear moduli, bulk moduli and the Pugh ratios of the stable and meta-stable Li/Na borohydrides under the Voigit-Reuss-Hill assumption, see **Figure 6**. Both the shear and bulk moduli of metal borohydrides are relatively small compared with metal oxides (such as Fe_2O_3 whose bulk and shear moduli are on the order of several hundreds of GPa). [98] The weaker mechanical strength is probably due to the rotational flexibility of large anions during the loading process. The shear moduli of Li and Na borohydrides are on the same order of magnitude of the Li and Na anodes respectively. According to Monroe and Newman's work, which is based on linear elasticity, the shear modulus is related to the ability to resist metal dendrite formation; additionally, [56, 57] the shear modulus of an electrolyte should be at least twice that of the lithium metal anode in order to prevent Li dendrite growth. [56, 58] The computed modulus of $Li_2B_{12}H_{12}$ is higher than that of $LiBH_4$ and is approximately equal to twice that of Li. This result suggests that the formation of such interphase may further help suppress dendrites. However, we must note that linear elasticity only provides qualitative bounds. Actually, contrary to the LLZO oxide, whose shear modulus is orders of magnitude higher than Li, the shear moduli of metal borohydrides are relatively close to twice that of the corresponding anodes. Thus, whether they are capable to suppress the dendrite formation needs careful experiments. We also calculated the Pugh ratio, an indicator of material ductility, of the metal borohydrides (materials are usually classified as ductile if the Pugh ratio is higher than 1.75 and brittle under 1.75). The values for Li borohydrides all exceed 1.75, indicating that they are likely ductile in nature. This is crucial for device level production which usually involves mechanical processing. [56] For Na, though some of the borohydrides could be classified as brittle, their Pugh ratio is a little below 1.75. We should stress again, that, while these results are encouraging because they suggest that mechanical processing is potentially facile, actual experimental validation is needed.

3.4 Implications on disordered phase transition and ionic conductivity

A high enough ionic conductivity is one of the most critical requirements for SSEs. In metal borohydrides, it is known that the cation transport is closely linked to anion rotations. [17 29, 52-54, 94] At low temperatures, the anions are arranged in an ordered manner and cation hopping is generally infrequent. At elevated temperatures, the anions can rotate and the rotational distortions are beneficial to the cation diffusion. Previous studies found that the Li/Na conductivities are orders of magnitude higher in the high-temperature anion disordered phases. Thus, one of the major challenges is to lower this transition temperature. Therefore, it is interesting to see whether there is any correlation between the phase transition temperatures and other structural properties of metal borohydrides. We calculated the anion rotation barrier of anions in different phases and the values are shown in **Figure 7 (a)** together with the experimentally observed phase transition temperatures. Despite the relatively small sample size, we found a positive correlation between the two variables (with Pearson correlation coefficient of 0.81, which is an indication of positive linear relationship), i.e., a larger rotation barrier is associated with a higher phase transition temperature. This is in agreement with previously observed thermally activated anion rotation process. [99, 100] In addition, the rotation barrier appears to be negatively correlated with the anion size, e.g., the BH_4^- containing phases have lower rotation barriers compared to other phases. Considering this, it might be interesting to see, experimentally, whether two metal borohydrides with different anion sizes could form solid solutions and whether such mixing will lower the phase transition temperatures without a major loss of ionic conductivity.

We also tried to gain some insights on cation diffusion based on DFT calculations and MD simulations. As mentioned earlier, ionic conductivity of a SE is relevant to the concentration of charge carriers and the intrinsic jump rates of charge carriers.[47 48] Usually, the second part could be estimated using transition-state based theories such as NEB calculations. However, as previously reported, since the cation hopping process is correlated with the anion

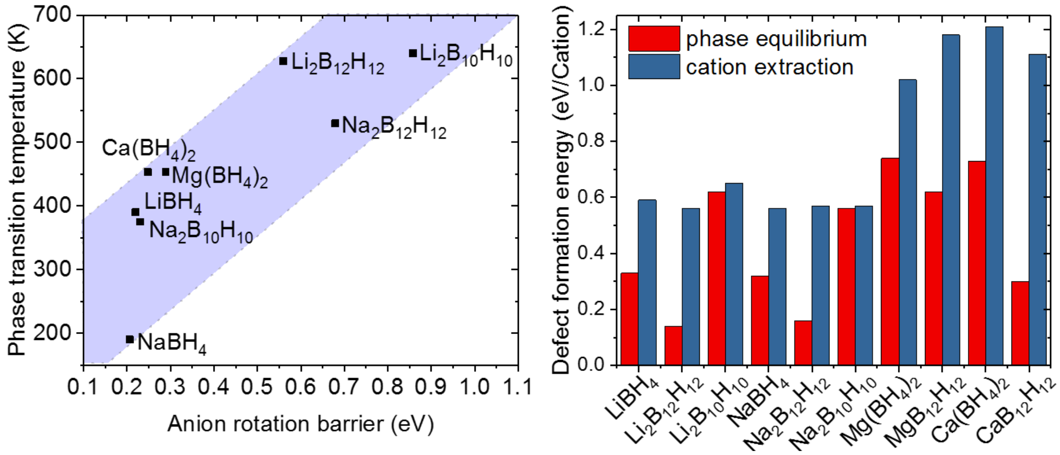

Figure 7. (a) Anion rotation barriers and experimental phase transition temperatures of metal borohydrides. The phase transition temperatures were derived from previously reported experimental data. [16, 17, 28] (b) The cation vacancy formation energies of metal borohydrides. The red bar shows the values obtained from phase equilibria and the blue one refers to the values from the cation extraction reaction. The detailed calculation details are shown in the Method section.

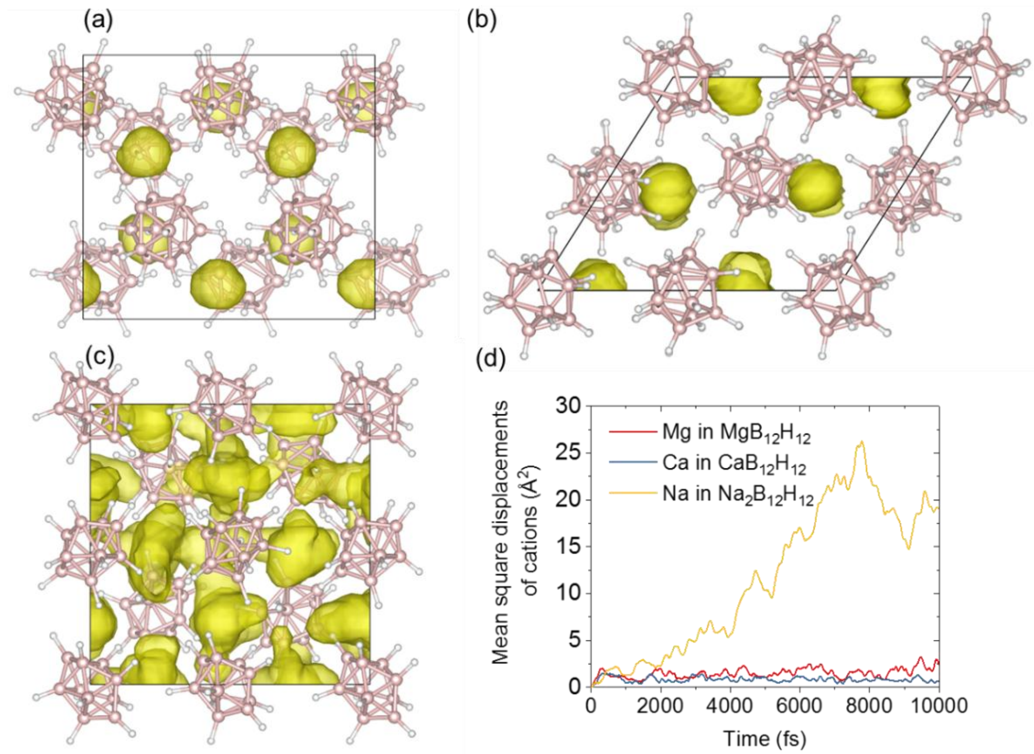

Figure 8. The cation nuclear density of (a) $CaB_{12}H_{12}$, (b) $MgB_{12}H_{12}$ and (c) $Na_2B_{10}H_{10}$ from 10 ps molecular dynamics simulations at 1000 K. The iso-surface values were set to 2×10^{-6} Å^{-3} fs^{-1}. The pink and white balls represent B and H atoms. The cations are omitted for clarity. (d) The mean square displacements of cations. It worth noting that we only show the first 10 picoseconds of MSDs so as to maintain the consistency of the results. The actual simulated timescale is slightly longer and main varies case by case, which is due to the wall-time of our computing facility. The detailed van Hove analysis could be found in **Figure S8.**

rotations, it is difficult to obtain the right cation hopping pathway. [101] [102] Alternatively, we evaluated the cation vacancy formation energies. Cationic vacancies are beneficial to the diffusion because they are the major charge carriers in most SEs. More cation vacancies indicate more vacant sites (or pathways) for the hopping process. **Figure 7 (b)** shows the cation vacancy formation energies of Li, Na, Ca and Mg borohydrides. Based on phase equilibria (as indicated by the red bars), the formation of cation vacancies in $B_{12}H_{12}^{2-}$-containing phases is more difficult than that in BH_4^--containing phases. This is probably related to the good thermodynamic stability of the former. Also, compared with divalent metal borohydrides (Mg

and Ca), monovalent metal cationic vacancies are easier to form, which is probably due to the larger electrostatic screening interaction. This indicates a lower concentration of available vacant sites and probably will lead to severe diffusion pathway blocking. We noticed here that, the values obtained using the phase equilibria involves the reaction of anion species, which is shown in Equation (2) in the method section. As mentioned earlier, such reactions may be kinetically sluggish. Thus, we also consider the direct extraction of cations from the crystal. (Equation (3) in the method section is an example.) The results are consistent with those obtained from the thermodynamics. Shown as blue bars in **Figure 7 (b)**, the formation energy of monovalent cationic vacancies are consistently smaller than divalent ones. These findings point to a low conductivity in divalent metal borohydrides. To confirm this speculation, we conducted *ab-initio* molecular dynamics simulations to study the intrinsic diffusion properties of Ca and Mg borohydrides. A comprehensive investigation on monovalent phases using this method can be found in a recent report by Brandon *et al.*[102] The nuclear densities of cations in $CaB_{12}H_{12}$ and $MgB_{12}H_{12}$ are shown in **Figure 8 (a)** and **(b)** respectively, in comparison with $Na_2B_{10}H_{10}$ from our previous study (c).[101] Despite the high simulation temperature (1000 K), Ca and Mg motions are localized whereas Na tend to diffuse frequently between neighboring sites. The diffusivity was quantified by the mean squared displacements in **Figure 8 (d)**. No obvious hopping of Ca and Mg was observed during the 10 ps time length. This implies that the diffusion of divalent metal cations is much slower than the monovalent ones. The van Hove correlation analysis also support this finding (see part 8 section 1 of SI). Based on the concentration and the intrinsic mobility of the carriers, Ca and Mg might not be suitable as solid electrolytes. Instead, they might be used as thin coating layers to protect the electrodes due to their relatively good electrochemical stability depending on their actual ionic conductivity. We want to note here that such a coating layer does not necessarily need to be highly conductive. [103] In the simulations, due to the relatively slow cation transport, statistically meaningful diffusion coefficient cannot be computed within the current simulation timescale. Actual conductivity needs to be measured experimentally.

4. Conclusions

We evaluated the phase stability, electrochemical stability, and mechanical properties of metal borohydrides as SEs for SSBs based on DFT calculations. Several experimentally observed cation conductive metal borohydrides were confirmed to be intrinsically stable. Under electrochemical oxidation, metal borohydrides are not thermodynamically stable and tend to be oxidized at relatively low voltages. However, the decomposition product may form a protective layer at the interface, preventing the electrolyte from further reactions and offering an improved electrochemical stability. The analysis of the mechanical properties computed using DFT indicates that metal borohydrides are ductile suggesting that facile mechanical processing is possible. However, their relatively small shear moduli suggest that they may not be stable against Li metal dendrite growth. Implications on phase transitions and cation diffusion properties were obtained by relating the experimental results to the DFT calculated descriptors. The phase transition temperature was found to be positively correlated with anion rotation barriers. For divalent Mg and Ca, both vacancy formation energy calculations and MD simulations indicate low diffusion constants.

Based on this study, we believe that metal borohydrides are potentially competitive solid electrolytes for Li and Na batteries. However, before they can be used in practical devices, two issues need to be addressed. First, the room temperature conductivity needs to be improved either by reducing the superionic phase transition temperature or by lifting the cation diffusivity of the low temperature phase. Second, these materials need to be characterized in real devices for a long number of cycles to rule out the effect of dendrites.

ASSOCIATED CONTENT

Supporting Information.
Detailed calculation methods; figures of molecular dynamics trajectories, decomposition products of borohydrites and grand potential phase diagrams; and tables of calculated entries, energies of metastable phases, elastic constants and formation energies of vacancies. This material is available free of charge via the Internet at http://pubs.acs.org.

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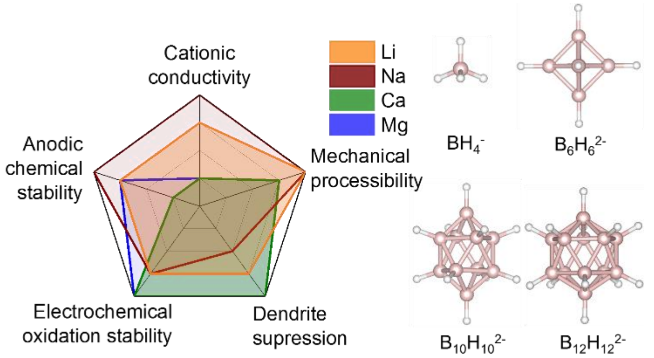
TOC


Cationic conductivity
Li
Na
Ca
Mg
Mechanical processibility
Anodic chemical stability
Electrochemical oxidation stability
Dendrite supression
BH_4^-
$B_6H_6^{2-}$
$B_{10}H_{10}^{2-}$
$B_{12}H_{12}^{2-}$